

Inventory & Supply Chain Pipeline

1. Storage Account for my Project

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The left sidebar shows the 'Overview' page for the storage account 'iaspstorage'. The main content area displays the 'Essentials' section with various properties and settings.

Essentials

Property	Value
Resource group	iasp-project-rg
Location	centralindia
Subscription	Azure for Students
Subscription ID	c070ba66-c7e0-4357-bcc3-a5da897ca6a7
Disk state	Available
Tags	Add tags
Performance	Standard
Replication	Locally redundant storage (LRS)
Account kind	StorageV2 (general purpose v2)
Provisioning state	Succeeded
Created	7/11/2026, 8:11:22 PM

Properties

Property	Value
Hierarchical namespace	Enabled
Default access tier	Hot
Blob anonymous access	Disabled
Blob soft delete	Disabled
Container soft delete	Disabled
Versioning	Unavailable due to migration
Change feed	Unavailable due to migration
NFS v3	Disabled
SFTP	Disabled
Storage tasks assignments	None

Security

Property	Value
Require secure transfer for REST API operations	Enabled
Storage account key access	Enabled
Minimum TLS version	Version 1.2
Infrastructure encryption	Disabled

Networking

Property	Value
Public network access	Enabled
Public network access scope	Enable from all networks
Private endpoint connections	0

2. Container for my Project (Supply Chain with all raw data)

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The left sidebar shows the 'Overview' page for the container 'iaspcontainer'. The main content area displays the 'Overview' section with various properties and settings.

Overview

Authentication method: Access key (Switch to Microsoft Entra user account)

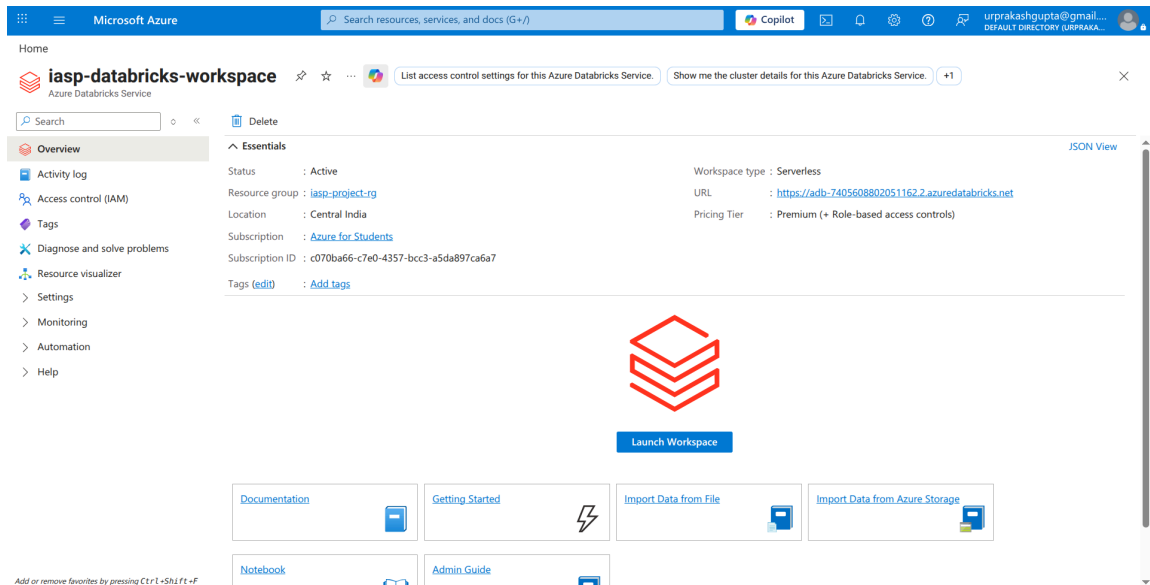
Search blobs by prefix (case-sensitive)

Only show active objects

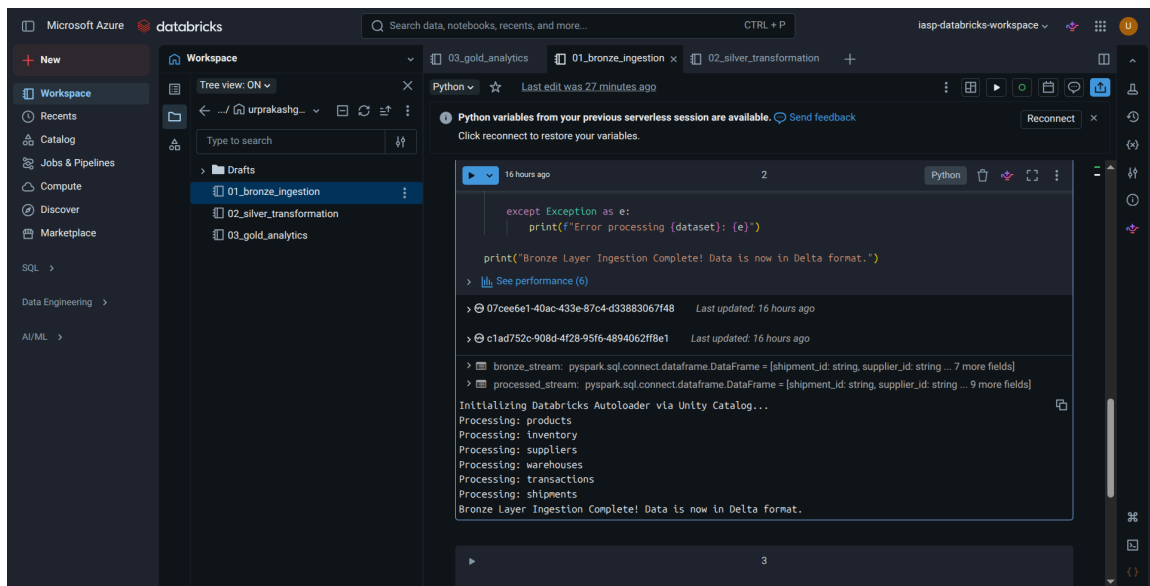
Showing all 4 items

Name	Last modified	Access tier	Blob type	Size	Lease state
bronze	7/11/2026, 8:14:19 PM				...
gold	7/11/2026, 8:14:33 PM				...
raw	7/11/2026, 8:13:58 PM				...
silver	7/11/2026, 8:14:28 PM				...

3. Azure Databricks workspace



4. The Bronze Ingestion (Databricks)



6. The Silver Transformation (Databricks)

Microsoft Azure databricks

Workspace

Tree view: ON

Type to search

Drafts

- 01_bronze_ingestion
- 02_silver_transformation
- 03_gold_analytics

Python

Last edit was 26 minutes ago

Python variables from your previous serverless session are available. Send feedback

Click reconnect to restore your variables.

Reconnect

```
clean_df: pyspark.sql.connect.dataframe.DataFrame = [shipment_id: string, supplier_id: string ... 12 more fields]
Initializing Silver Layer Transformations (SCD Type 2)...
Cleaning and processing SCD2 for: products (Key: product_id)
Cleaning and processing SCD2 for: inventory (Key: inventory_id)
Cleaning and processing SCD2 for: suppliers (Key: supplier_id)
Cleaning and processing SCD2 for: warehouses (Key: warehouse_id)
Cleaning and processing SCD2 for: transactions (Key: transaction_id)
Cleaning and processing SCD2 for: shipments (Key: shipment_id)
Silver Layer Cleaning & SCD2 Implementation Complete!
```

16 hours ago (11s)

3

SQL

```
%sql
SELECT * FROM silver_products LIMIT 5;
```

See performance (1)

Optimize

_sqldf: pyspark.sql.connect.dataframe.DataFrame = [product_id: string, product_name: string ... 17 more fields]

	product_id	product_name	category	brand	cost_price	set
1	P1157	Stay	Clothing>Men>Topwear	Dell	48503.95	21513
2	P1215	Audience	Electronics>Mobile>Accessori...	Boat	43487.03	2851.1
3	P1010	Decision	Electronics>Mobile>Accessori...	Boat	28434.89	63480
4	P1279	Open	Electronics>Mobile>Accessori...	Boat	35897.19	51618

7. The Gold KPIs (Databricks)

Microsoft Azure databricks

Workspace

Tree view: ON

Type to search

Drafts

- 01_bronze_ingestion
- 02_silver_transformation
- 03_gold_analytics

Python

Last edit was 26 minutes ago

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Click reconnect to restore your variables.

Reconnect

```
inventory: pyspark.sql.connect.dataframe.DataFrame = [inventory_id: string, product_id: string ... 13 more fields]
low_stock: pyspark.sql.connect.dataframe.DataFrame = [product_id: string, product_name: string ... 3 more fields]
product_movement: pyspark.sql.connect.dataframe.DataFrame = [product_id: string, total_units_moved: double ... 3 more fields]
products: pyspark.sql.connect.dataframe.DataFrame = [product_id: string, product_name: string ... 19 more fields]
sales_summary: pyspark.sql.connect.dataframe.DataFrame = [category: string, category_revenue: double ... 1 more field]
sales_trends: pyspark.sql.connect.dataframe.DataFrame = [sales_date: timestamp, daily_revenue: double ... 1 more field]
transactions: pyspark.sql.connect.dataframe.DataFrame = [transaction_id: string, product_id: string ... 15 more fields]
Expanding Gold Layer with Business KPIs...
Generating Low Stock Alerts...
Generating Product Movement...
Generating Sales Summary...
Generating Sales Trends...
Gold Layer KPI Expansion Complete! The data is now fully ready for Power BI.
```

16 hours ago (1s)

4

SQL

```
%sql
SELECT * FROM gold_inventory_summary ORDER BY total_damaged_stock DESC;
```

See performance (1)

Optimize

Table

8. Gold inventory summary

Microsoft Azure databricks

Search data, notebooks, recents, and more... CTRL + P

Workspace

Tree view: ON

Type to search

Drafts

- 01_bronze_ingestion
- 02_silver_transformation
- 03_gold_analytics

Python

Last edit was 26 minutes ago

Python variables from your previous serverless session are available. Send feedback

Click reconnect to restore your variables.

Reconnect

16 hours ago (9s) 4

SQL

```
%sql
SELECT * FROM gold_inventory_summary ORDER BY total_damaged_stock DESC;
```

See performance (1)

Optimize

	product_id	total_stock_available	total_damaged_stock
1	P1472	51223	1057
2	P1101	42829	988
3	P1181	33419	957
4	P1425	25684	881
5	P1219	20955	820
6	P1336	42569	813
7	P1304	19887	810
8	P1125	31712	807
9	P1403	43308	795
10	P1324	32459	793
11	P1315	20414	779
12	P1020	42421	764
13	P1406	17836	763
14	P1271	38312	755

Microsoft Azure databricks

Search data, notebooks, recents, and more... CTRL + P

Workspace

Tree view: ON

Type to search

Drafts

- 01_bronze_ingestion
- 02_silver_transformation
- 03_gold_analytics

Python

Last edit was 26 minutes ago

Python variables from your previous serverless session are available. Send feedback

Click reconnect to restore your variables.

Reconnect

16 hours ago (2s) 5

SQL

```
%sql
SELECT * FROM gold_supplier_performance ORDER BY delayed_shipments DESC;
```

See performance (1)

Optimize

_sqldf: pyspark.sql.connect.dataframe.DataFrame = [supplier_id: string, total_shipments: long ... 1 more field]

	supplier_id	total_shipments	delayed_shipments
1	S202	271	230
2	S217	260	229
3	S201	283	226
4	S212	261	223
5	S211	258	221
6	S219	259	217
7	S210	257	217
8	S215	249	216
9	S200	253	213
10	S203	248	212
11	S214	251	212
12	S206	249	209